SS1 MATHEMATICS – THIRD TERM

WEEK 2: MENSURATION II (AREA OF SECTORS, SEGMENTS & CONES)

1. INTRODUCTION (Short Note)

Last week focused on **arc length and sectors**.
This week focuses on **AREA**:

- Area of **sector** (slice of a circle)
- Area of **segment** (part cut by a chord)
- Relationship between **sector and cone**

2. AREA OF A SECTOR

Concept

A sector is a **fraction of a circle**, so its area is a fraction of the total area.

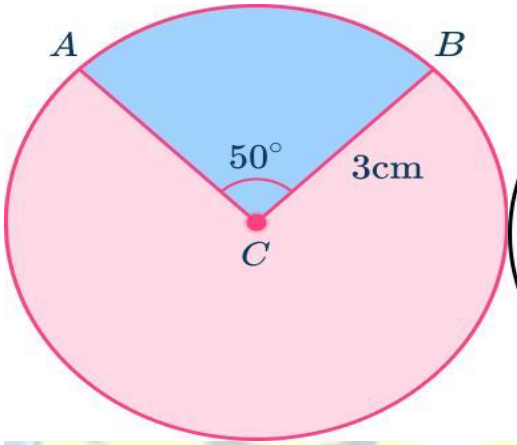
Formula

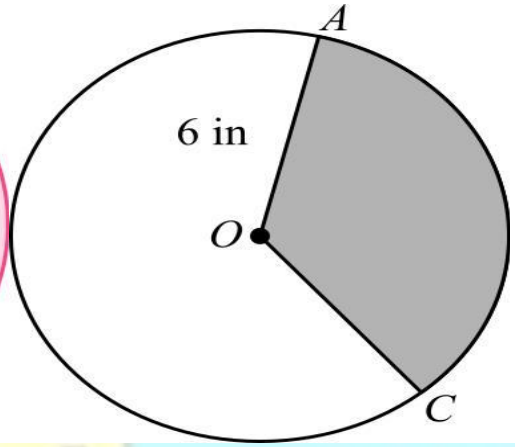
Area of Sector = $(\theta / 360) \times \pi r^2$

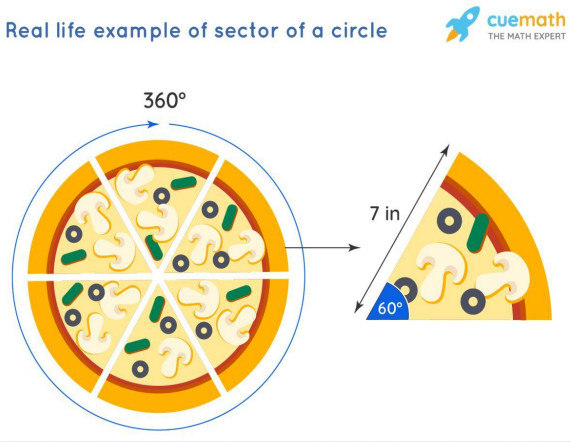
Where:

- θ = angle at center
- r = radius

Visual Idea







Example 1

r = 10 cm, θ = 135°, π = 3.142

Area = (135 / 360) × 3.142 × 10^2
= (3 / 8) × 3.142 × 100
= 0.375 × 314.2
= **117.825 cm^2 ≈ 117.83 cm^2**

3. AREA OF A SEGMENT

Concept

Segment = Sector − Triangle

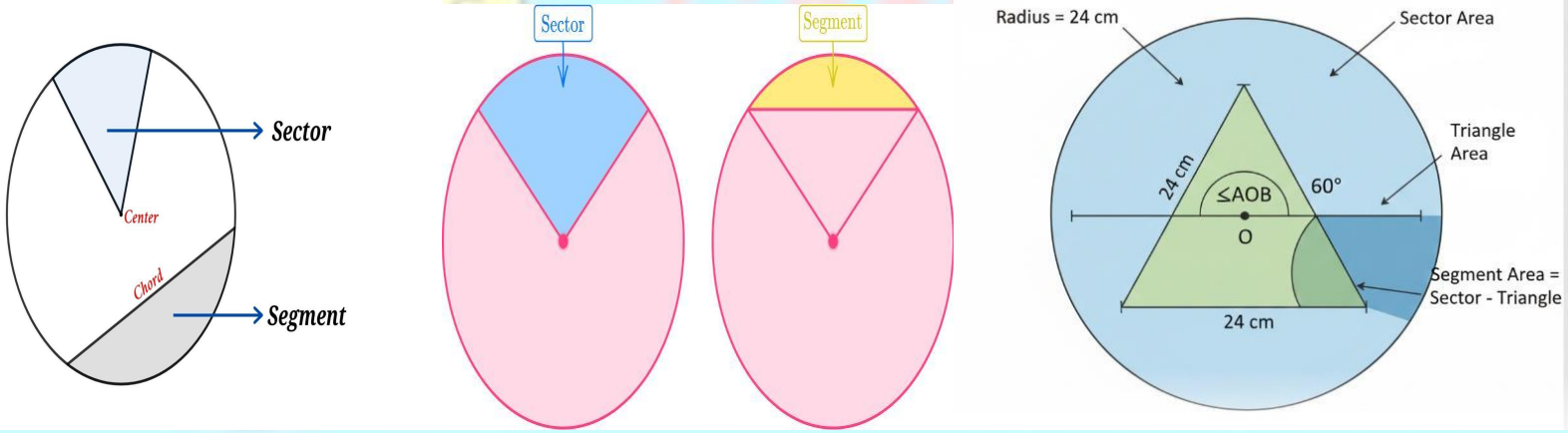
Formula (WPS Format)

Area of Segment = $(\theta / 360) \times \pi r^2 - (1/2 \times r^2 \times \sin\theta)$

Explanation

- First find **sector area**
- Then subtract **triangle area**

Visual Breakdown



Example 2

$r = 14$ cm, $\theta = 90°$, $\pi = 22/7$

Step 1: Sector Area
Area = $(90 / 360) \times 22/7 \times 14^2$
$= (1 / 4) \times 22/7 \times 196$
$= 154$ cm²

Step 2: Triangle Area
Area = $1/2 \times 14^2 \times \sin 90°$
$= 1/2 \times 196 \times 1$
$= 98\ cm^2$

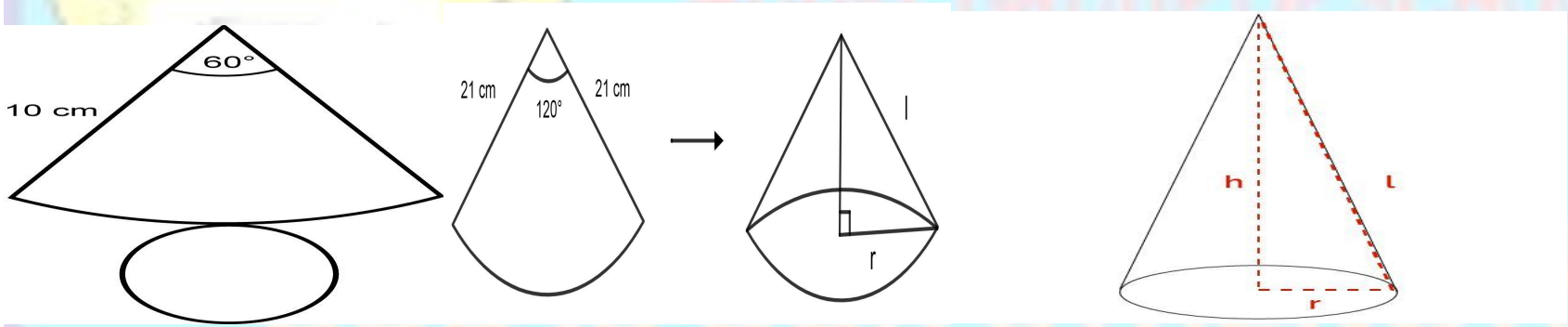
Step 3: Segment Area
Segment = 154 − 98
= **56 cm**

4. RELATIONSHIP BETWEEN SECTOR AND CONE

Key Idea

A **sector can be folded into a cone**

Transformation



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Important Relationships

Sector (Flat)	Cone (3D)
Radius (r)	Slant height (l)
Arc length	Base circumference
Sector area	Curved surface area

Key Formulas (WPS Format)

- Arc Length = $(\theta / 360) \times 2\pi r$
- Base circumference = $2\pi R$

So:
$(\theta / 360) \times 2\pi r = 2\pi R$

Simplifies to:
$(\theta / 360) \times r = R$

Curved Surface Area of Cone

$CSA = \pi R l$

5. EXAMPLE (SECTOR → CONE)

Given:
$r = 15$ cm, $\theta = 216°$

(a) Find Base Radius (R)

$(\theta / 360) \times r = R$

$(216 / 360) \times 15 = R$
$= (3 / 5) \times 15$
$=$ **9 cm**

(b) Find Height (h)

Using Pythagoras:

$l^2 = R^2 + h^2$

$15^2 = 9^2 + h^2$
$225 = 81 + h^2$
$h^2 = 144$
$h =$ **12 cm**

(c) CSA of Cone

$CSA = \pi Rl$
$= \pi \times 9 \times 15$
$=$ **135π cm²**

6. KEY POINTS TO REMEMBER

- Sector area = fraction of circle
- Segment = sector − triangle
- $\sin 90° = 1$ (very important)
- Sector → Cone transformation:
    - radius → slant height
    - arc → circumference
- CSA of cone $= \pi Rl$

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7. REVISION QUESTIONS

1. What is the formula for area of a circle?
2. Write formula for sector area
3. How do you find segment area?
4. State triangle area formula using sine
5. What happens to arc length when folded into cone?
6. What does sector radius become?

8. EVALUATION (CLASSWORK)

1. Write segment formula
2. Sector with angle 60° is what fraction?
3. Write CSA formula of cone
4. Find sector area: $r = 21$ cm, $\theta = 60°$
5. Write relation between θ, r, R

9. ASSIGNMENT

1. $r = 10$ cm, $\theta = 72°$
   (a) Arc length
   (b) Sector area
   (c) Perimeter
2. Find segment area:
   $r = 8$ cm, $\theta = 60°$
3. Sector → Cone:
   $r = 25$ cm, $\theta = 144°$
   (a) Base radius
   (b) Height
   (c) CSA

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